

LLM 5-Days Report

LLM lardan so'ralayotgan narsalar
query deb ataladi.

Arxitektura 3 ta - BLLM dan iborat
Block Foundations,
Transformer,
Training va Performance.

Embedding - berilgan discrete
categorical datani $\rightarrow$ low dimension
number vektorga o'tkazish

Attention - so'zlarni bir-biriga
bog'laydigan tizimish jarayoni

Demak LLM Arxitekturasida eng
boshchisi Block Foundation hisoblanadi
 $\rightarrow$ Feed Forward Layers (FFN)

Layers - bu matematik amallar
tizimi xisoblanadi va inputni
outputga o'tkazadi

$$\text{Output} = \text{Activation} (\text{Input} \times W)$$

LLM da NN da layerlar 2 qism
ga bo'linadi

1 qism: Attention Layer (Context Engine)

transformer tokenlar bor - boru b-
gaplashadi va ular orasidagi
bog'lanishlarni xoslaydi

2-qismli Feed Forward Network
layer (Knowledge Engine)

transformer soxlar alohida ishlaydi
tamirlaydi

Attention layer $\Rightarrow$ Grouped Query
Attention (GQA)

FFN layer $\Rightarrow$ SwiGLU
(Gated Linear Units)

LLM Arxitekturasining 4 bosqichi

1-Bosqich Tokenizatsiya va embedding

2-Bosqich Attention layer (mulqot
va bog'lanish)

3-Bosqich FFN layer (fakti qadrlar
topash)

4-Bosqich Birlashtirish (jarebini tayyorlash)

LLM da layerlar joylashuvi
2 xil bor

1- Serial

2- Parallel

Bo'sda ML da ensemble algorithmlar
bor stacking, Voting, Bagging
Boosting bularda ham shlar
serial va parallel bo

Block Foundation 2. bo'sqichi Normalisation.

Normalisation bu scalingga o'sha
klassifitsiya NN ichidagi ragamlarni
kichik sonlarda ushlab qurish.
($x/255$)

Normalisationni 2 xil usul
ma'rufl.

① Layer Norm (GPT2, GPT3-) eshik
mean variance bilan shlar
uham bor safar GPU qoshuncha
xotira evasga shlaydi

② RMS Norm (Root mean square
Normalisation) - samonaviy usul.
xor bor so'zni qayind shlar
headrotok (RMSE) ga b'olib shlaydi.

Block Foundationda Aktivatsion Funksion.

bor lineartiyalar - non lineartiyalar
tamoy chunki bo'sda 2D emas 3D
bor bor u no chugur shlar shlar

Reasoning - bu Activation F
ning berovdigan gismat, 11 mni
yaxshi yanvarlogon kirsatuvchi
malumot xosoblanadi.

Activation Function - input eifa
vada kongan gars gismat kiyin
bosqichga o'tishi uchun blocklarni
(nolga tenglamish, sarsab yuborish)
lanavlogon belgilab beradi.

Assosy Turlard

RELU / GELU / Swig GU / GEGLU

Transform Architecture

Tokenization - sozni hechik qism
larga ajratish jarayoni.

Maksadi tuzrog training, tuzrog
larga javab berishi, androg
ajratilrog belvishi uchun filit

Pre Norm va Post Norm

Pre Norm - Input $\Rightarrow$ Tokenization $\Rightarrow$
 $\Rightarrow$ Normalization $\Rightarrow$ Attention/FPN $\Rightarrow$
 $\Rightarrow$ Activation.

Post Norm - Input $\Rightarrow$ Tokenization $\Rightarrow$
 $\Rightarrow$ Attention/FPN $\Rightarrow$ Normalization $\Rightarrow$
 $\Rightarrow$ Activation.

Pre Norm samonardy usul xisob-
k

Architecture Variants! ~~Mbammo~~

Modellova juda usun taplar bilan
Hilaganda, GPU xotira cache
(KV - key value xotira ombori) bilan
xisob ketadi $\rightarrow$ model sekinlashadi
 $\rightarrow$ GPU out of memory bo'ladi.
Yechim: Architecture Variants
mavzu:

MHA - (Multi-Head Attention)

MQA - (Multi-Query Attention)

GQA - (Grouped-Query Attention)

LLM da son, qayon o'lgan
tushuncha yig' fargat or yoki
top o'lgan narsa bor.

MHA - har bir query dagi
suzalning ma'nosi (grammatika,
positive / Negative, history...) faron
barkon topish uchun har bir
qisimga alohida key va value.

MQA - barcha qisimga bitta
key va value beradi.

GQA - qismlar malum qisimga solinadi
va har bir qisim u alohida key va
value beradi.